

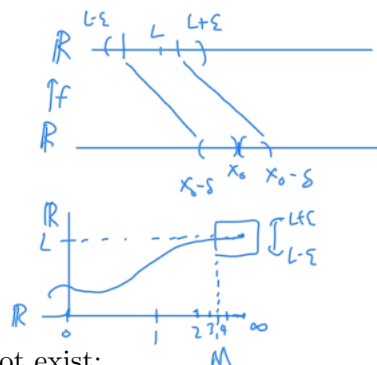
1 Review of limits

Feedback welcomed :)

Question 1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function and $x_0 \in \mathbb{R}$ be a point. State the definition of $\lim_{x \rightarrow x_0} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$.

$L = \lim_{x \rightarrow x_0} f(x)$ means for all $\epsilon > 0$, there exists $\delta > 0$ such that $|f(x) - L| < \epsilon$ for all $x \neq x_0$ with $|x - x_0| < \delta$.

$L = \lim_{x \rightarrow \infty} f(x)$ means for all $\epsilon > 0$, there exists M such that $|f(x) - L| < \epsilon$ for all $x \geq M$.



Question 2. Find the following limits or determine that they do not exist:

a) $\lim_{x \rightarrow \infty} \left(5 + \frac{1}{x} \right) = \lim_{x \rightarrow \infty} 5 + \lim_{x \rightarrow \infty} \frac{1}{x} = 5$

b) $\lim_{x \rightarrow \infty} \frac{\sqrt{5x^2 - 3}}{3x + 2} \rightarrow \frac{\sqrt{5 - 3/x^2}}{3 + 2/x} \rightarrow \frac{\sqrt{5}}{3}$

c) $\lim_{x \rightarrow \pi/2} \frac{11(\tan x)^2 + x}{2(\tan x)^2 + 1} = \lim_{y \rightarrow \infty} \frac{11y^2 + \tan^{-1}y}{2y^2 + 1} = \frac{11}{2}$
($\lim_{y \rightarrow \infty} \tan^{-1}y = \frac{\pi}{2}$)

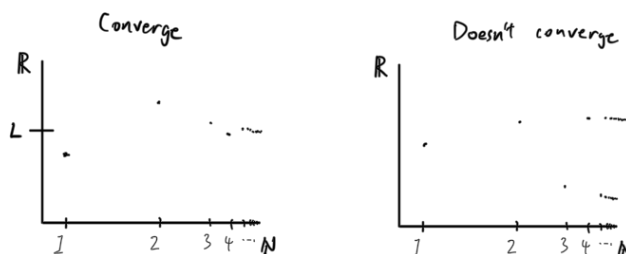
d) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$

e) $\lim_{x \rightarrow \infty} x \sin(1/x) = \lim_{x \rightarrow \infty} \frac{\sin(1/x)}{1/x} = \lim_{y \rightarrow 0} \frac{\sin y}{y} = 1$
L.H. is another solution

f) $\lim_{x \rightarrow \infty} \frac{e^{x^2}}{xe^x} \rightarrow \frac{e^{x^2-x}}{x} \xrightarrow{\text{L.H.}} \frac{(2x-1)e^{x^2-x}}{1} \rightarrow \infty$
L.H. won't work

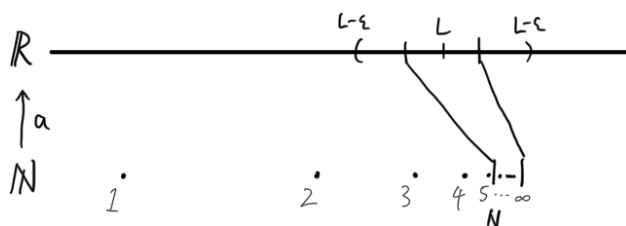
2 Sequential limits

A sequence a_n converges if as n gets large, a_n gets closer and closer to a real number L .



The formal definition of sequential limits is as follows:

$\lim_{n \rightarrow \infty} a_n = L$ if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $|a_n - L| < \epsilon$ for all $n \geq N$.



The tools we developed for limits of functions can be used to find limits of sequences thanks to the theorem:

If $\lim_{x \rightarrow \infty} f(x) = L$, then $\lim_{n \rightarrow \infty} a_n = L$ where $a_n = f(n)$.

Question 3. Which of the following sequences converge? If they do, find the limit.

a) $a_n = \frac{3n+2}{n^2+1} \rightarrow \frac{3+\frac{2}{n}}{n+1} \rightarrow 0$

b) $a_n = \frac{(n+1)! \cdot n^2}{(n+3)!} \rightarrow \frac{n^2}{(n+2)(n+3)} \rightarrow 1$

c) $a_n = (-1)^n$ $-1, 1, -1, 1$ doesn't converge

d) $a_n = \cos(2\pi n)$ $1, 1, 1, 1 \rightarrow 1$

e) $a_1 = 1$ and $a_{n+1} = a_n + \left(\frac{1}{n+1} - \frac{1}{n}\right)$ $a_n = \frac{1}{n} \rightarrow 0$

f) $a_n = \frac{4^n}{n}$ $\frac{4^x}{x} \xrightarrow{L.H.} \frac{\ln 4 \cdot 4^x}{1} \rightarrow \infty$

g) $a_n = \frac{\ln n}{n}$ $\frac{\ln x}{x} \xrightarrow{L.H.} \frac{1/x}{1} \rightarrow 0$

h) $a_n = \frac{\ln n}{n^{0.01}}$ $\frac{\ln x}{x^{0.01}} \xrightarrow{L.H.} \frac{1/x}{0.01 x^{0.01-1}} = \frac{1}{0.01 x^{0.99}} \rightarrow 0$

i) $a_n = (n+1) \sin(1/n)$ $n \sin(1/n) \rightarrow 1$ (see 2e)
 $\sin(1/n) \rightarrow 0$

j) $a_n = n \ln\left(1 + \frac{1}{n}\right)$ $\frac{\ln(1+\frac{1}{n})}{1/n} \rightarrow \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} \xrightarrow{L.H.} \frac{1}{1+x} \rightarrow 1$

k) $a_1 = 1$ and $a_{n+1} = \frac{5+a_n}{2} \rightarrow 5$
 $1, 3, 4, 4.5, 4.75, \dots$ $(5-a_{n+1}) = \frac{(5-a_n)}{2} \Rightarrow 5-a_n = \frac{5-a_1}{2^{n-1}} = \frac{1}{2^{n-3}} \rightarrow 0$

l) $a_n = \frac{n^{100}}{2^n}$ $\xrightarrow{L.H. \times 100} \frac{100!}{(\ln 2)^{100} 2^x} \rightarrow 0$

Alternatively note: $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$

3 Continuous function theorem

Question 4. Argue from definitions that if $\lim_{n \rightarrow \infty} a_n = L$ and if f is a function that is continuous at L (i.e., $\lim_{x \rightarrow L} f(x) = f(L)$), then $\lim_{n \rightarrow \infty} f(a_n) = f(L)$.

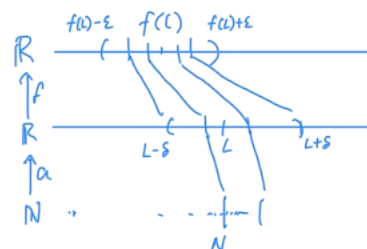
For all $\epsilon > 0$, there exists $\delta > 0$ (by continuity) with

$$|x - L| < \delta \Rightarrow |f(x) - f(L)| < \epsilon,$$

and then exists $N \in \mathbb{N}$ (by convergence) with

$$n \geq N \Rightarrow |a_n - L| < \delta.$$

$$\text{Thus, } n \geq N \Rightarrow |f(a_n) - f(L)| < \epsilon.$$



Question 5. Find the limits of the following sequences or determine that they do not exist:

a) $\lim_{n \rightarrow \infty} \frac{\sqrt{n+1}}{\sqrt{n+3}} = \sqrt{\lim_{n \rightarrow \infty} \frac{n+1}{n+3}} = \sqrt{1} = 1$

b) $\lim_{n \rightarrow \infty} (\ln n - \ln(n+1)) = \lim_{n \rightarrow \infty} \ln\left(\frac{n}{n+1}\right) = \ln\left(\lim_{n \rightarrow \infty} \frac{n}{n+1}\right) = \ln 1 = 0$

c) $\lim_{n \rightarrow \infty} \sqrt[n]{n^2+1} \xrightarrow{L.H.} \frac{2n}{n^2+1} \rightarrow 0 \rightarrow e^0 = 1$
 $e^{\ln \sqrt[n]{n^2+1}} = e^{\frac{1}{n} \ln(n^2+1)}$

d) $\lim_{n \rightarrow \infty} \left(\frac{n}{n+x}\right)^n$ (any positive real x)

$$= e^{n \ln \frac{n}{n+x}} = e^{-\frac{n \ln(1+\frac{x}{n})}{1/n}} \xrightarrow{(*)} \lim_{n \rightarrow \infty} \frac{\ln(1+\frac{x}{n})}{1/n} = \lim_{y \rightarrow 0} \frac{\ln(1+y)}{y} = 1$$

Alternatively note $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$

4 Sandwich theorem

Sandwich theorem states that if $a_n \leq b_n \leq c_n$ for all $n \geq n_0$, then (assuming the limits exist)

$$\lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n \leq \lim_{n \rightarrow \infty} c_n.$$

Question 6. Find the limits of the following sequences or determine that they do not exist:

a) $\lim_{n \rightarrow \infty} \frac{\cos(n)}{n} - \frac{1}{n} \leq \frac{\cos(n)}{n} \leq \frac{1}{n} \rightarrow 0$

b) $\lim_{n \rightarrow \infty} \frac{(-1)^n \cdot n}{1+n}$ doesn't converge. Sandwich theorem is inconclusive because $\frac{-n}{1+n} \leq \frac{(-1)^n \cdot n}{1+n} \leq \frac{n}{1+n}$

c) $\lim_{n \rightarrow \infty} \frac{3^n}{n!}$ $0 \leq \frac{3^n}{n!} = \frac{3}{n!} \leq \frac{3}{n \cdot (n-1) \cdots 2 \cdot 1} \leq \frac{27}{2^n} \rightarrow 0$

d) $\lim_{n \rightarrow \infty} \frac{n!}{n^n}$ (hint: compare with $1/n$)

e) $\lim_{n \rightarrow \infty} \frac{2^{(n+1)n}}{n!}$ (hint: $2^{(n+1)n} = 4^n 4^{n-1} \cdots 4^2 4^1$)

$$0 \leq \frac{n!}{n^n} = \frac{n \cdot (n-1) \cdots 2 \cdot 1}{n \cdot n \cdots n} \leq \frac{1}{n} \rightarrow 0$$

$$\frac{2^{(n+1)n}}{n!} = \frac{4^n \cdot 4^{n-1} \cdots 4^2 \cdot 4^1}{n \cdot (n-1) \cdots 2 \cdot 1} \geq \frac{4^n}{n} \rightarrow \infty$$